

Gurugram University
Scheme of Studies and Examination
Bachelor of Technology (Civil Engineering)
Semester-IV

Sr. No.	Course Code	Course Title	Hours per week	Contact hours per week	Credit	Examination Schedule (Marks)				Duration of Exam (Hours)
			L-T-P			Class work	Theory	Practical	Total	
1.		Organization Behavior	3-0-0	3	3	30	70	-	100	3
2.		Fluid Mechanics-II	3-0-0	3	3	30	70	-	100	3
3.		Design of concrete structure	3-0-0	3	3	30	70	-	100	3
4.		Structural Analysis	3-0-0	3	3	30	70	-	100	3
5.		Surveying-II	3-0-0	3	3	30	70	-	100	3
6.		Material Testing and Evaluation	3-0-0	3	3	30	70	-	100	3
7.		Fluid Mechanics-II lab	0-0-2	2	1	50	-	50	100	3
8.		Structural Analysis Lab	0-0-2	2	1	50	-	50	100	3
9.		Surveying-II Lab.	0-0-2	2	1	50	-	50	100	3
10.		Material Testing & Evaluation Lab.	0-0-2	2	1	50	-	50	100	3
11.		RCC Drawing Lab.	0-0-2	2	1	50	-	50	100	3
TOTAL					23				1100	

Note:

- Students will be allowed to use non-programmable scientific calculator. However, sharing of Calculator and other materials will not be permitted in the examination.
- (A) each student has to undergo practical training of 4/6 weeks in an Industry/ Institute/ Professional Organization/ Research Laboratory/ training centre etc and its evaluation shall be carried out in the V semester on the basis of seminar, viva-voce, report and certificate of practical training obtained by the student.

Course Name	:	ORGANIZATIONAL BEHAVIOUR	
Course Code	:		External marks: 70
Credits	:	3	Internal marks: 30
L-T-P	:	3-0-0	Total marks: 100

Course Objectives:

The objective of this course is to expose the students to basic concepts of management and provide insights necessary to understand behavioral processes at individual, team and organizational level.

COURSE CONTENT

UNIT – 1

Introduction of Management- Meaning, definitions, nature of management; Managerial levels, skills and roles in an organization; Functions of Management: Planning, Organizing, staffing, Directing & Controlling, Interrelationship of managerial functions, scope of management & Importance of management. Management and social responsibility, difference between management and administration.

UNIT – 2

Introduction of organization:-

Meaning and process of Organization, Management v/s Organization;

Fundamentals of Organizational Behavior: Concepts, evolution, importance and relationship with other Fields; Contemporary challenges and opportunities of OB.

Individual Processes and Behavior-Personality- Concept, determinants and applications; **Perception-** Concept, process and applications, **Learning-** Concept ,theories ; **Motivation-** Concept, techniques and importance

UNIT - 3

Interpersonal Processes- Teams and Groups- Definition of Group, Stages of group development, Types of groups, meaning of team, merits and demerits of team; difference between team and group, **Conflict-** Concept, sources, types, management of conflict; **Leadership:** Concept, function, styles & qualities of leadership.

Communication – Meaning, process, channels of communication, importance ,barriers and overcome of communication..

UNIT - 4

Organizational Processes: Organizational structure - Meaning and types of organizational structure and their effect on human behavior; **Organizational culture** Elements, types and factors affecting organizational culture. **Organizational change:** Concept, types & factors affecting organizational change, Resistance to Change.

Course Outcomes: By the end of this course the student will be able to:

1. Students will be able to apply the managerial concepts in practical life.
2. The students will be able to understand the concept of organizational behavior at individual level and interpersonal level.
3. Students will be able to understand the behavioral dynamics in organizations.
4. Students will be able to understand the organizational culture and change

Suggested Books:

1. Robbins, S.P. and Decenzo, D.A. Fundamentals of Management, Pearson Education Asia, New Delhi.
2. Stoner, J et. al, Management, New Delhi, PHI, New Delhi.
3. Satya Raju, Management – Text & Cases, PHI, New Delhi.
4. Kavita Singh, Organisational Behaviour: Text and cases. New Delhi: Pearson Education.
5. Pareek, Udai, Understanding Organisational Behaviour, Oxford University Press, New Delhi.
6. Robbins, S.P. & Judge, T.A., Organisational Behaviour, Prentice Hall of India, New Delhi.
7. Ghuman Karminder, Aswathappa K., Management concept practice and cases, Mc Graw Hill education.
8. Chhabra T. N., Fundamental of Management, Sun India Publications-New Delhi.

Fluid Mechanics-II			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	3-0-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To introduce the students to various hydraulic engineering problems like laminar flow, open channel flows, flow through pipes, hydraulic jump and its applications.
- At the completion of the course, the student should be able to relate the theory and practice of problems in hydraulic engineering

COURSE CONTENT

UNIT A

Module 1: Laminar Flow

Laminar flow through: circular pipes, annulus and parallel plates. Stoke's law, Measurement of viscosity.

Module 2: Turbulent Flow

Reynolds experiment, Transition from laminar to turbulent flow, Definition of turbulence, scale and intensity, Causes of turbulence, instability, mechanism of turbulence and effect of turbulent flow in pipes. Reynolds stresses, semi-empirical theories of turbulence, Prandtl's mixing length theory, universal velocity distribution equation. Resistance to flow of fluid in smooth and rough pipes

UNIT B

Module 3: Flow through Pipes

Loss of head through pipes, Darcy-Wiesbatch equation, minor losses, total energy equation, hydraulic gradient line, Pipes in series, equivalent pipes, pipes in parallel, siphon, power transmission through pipes, Analysis of pipe networks: water hammer in pipes and control measures, branching of pipes.

UNIT C

Module 4: Open Channel Flow: Uniform flow

Definition, Comparison between open channel flow and pipe flow, geometrical parameters of a channel, classification of open channel flow.

Uniform Flow- Continuity Equation, Energy Equation and Momentum Equation, Characteristics of uniform flow, Chezy's formula, Manning's formula. Factors affecting Manning's Roughness Coefficient 'n', Most economical section of channel, Computation of Uniform flow, Normal depth.

Module 5: Open Channel Flow: Non-Uniform Flow

Specific energy, Specific energy curve, critical flow, discharge curve, Specific force, Specific depth, and Critical depth. Channel Transitions, Gradually Varied Flow- Dynamic Equation of Gradually Varied Flow, Classification of channel bottom slopes, Classification of surface profile

UNIT D

Module 6: Hydraulic Jump

Theory of hydraulic jump, Elements and characteristics of hydraulic jump in a rectangular Channel, length and height of jump, location of jump, Types, applications and location of hydraulic jump. Energy dissipation and other uses, surge as a moving hydraulic jump,

Module 7:

Surges, Positive and negative surges, Dynamics of Fluid Flow- Momentum principle, applications: Force on plates, pipe bends, moments of momentum equation,

COURSE OUTCOMES:

- The students will be able to apply their knowledge of fluid mechanics in addressing problems in open channels.
- They will possess the skills to solve problems in uniform, gradually and rapidly varied flows in steady state conditions.
- They will have knowledge in hydraulic jump and its applications.

SUGGESTED BOOKS:

- Hydraulics and Fluid Mechanics, P.M. Modi and S.M. Seth,
- Theory and Applications of Fluid Mechanics, K. Subramanya, Tata McGraw Hill.
- Open channel Flow, K. Subramanya, Tata McGraw Hill.
- Open Channel Hydraulics, Ven Te Chow, Tata McGraw Hill.
- Burnside, C.D., "Electromagnetic Distance Measurement," Beekman Publishers, 1971
- Fluid Mechanic and Hydraulic machines R.K. BANSAL

DESIGN OF CONCRETE STRUCTURE			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	3-0-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- The aim of this course is to provide students with a thorough understanding of the design of reinforced concrete structures.
- To become familiar with professional and contemporary issues in the design and fabrication of reinforced concrete members.
- Be able to identify and interpret the appropriate relevant industry design codes.
- The course focuses on understanding the behaviour of reinforced concrete components and systems subjected to gravity as well as lateral loads.
- Topics covered will include: design of beams, Column and slabs, detailing of reinforcement, design of foundation and retaining wall.

COURSE CONTENT

UNIT-A

Module 1: Design methodology in Reinforced Concrete & Working stress Method

Working stress and limit state methods, Limit state v/s working stress method, Building codes, Normal distribution curve, Characteristic strength and Characteristics loads, Design values, Partial safety factors and Factored loads, Stress-Strain relationship for concrete and steel. Working Stress Method: Basic assumptions, permissible stresses in concrete and steel, design of singly and doubly reinforced rectangular and flanged beams in flexure, steel beam theory, inverted flanged beams, design examples.

Module 2: Limit State Method

Basic assumptions, Analysis and design of singly and doubly reinforced rectangular flanged beams, minimum and maximum reinforcement requirement and design examples. Continuous Beams both method -Basic assumptions, Moment of inertia, settlements, Modification of moments, maximum moments and shear.

UNIT-B

Module 3: Concrete Reinforcement and Detailing

Requirements of good detailing, Cover to reinforcement, Spacing of reinforcement, Reinforcement Splicing, Anchoring reinforcing bars in flexure and shear, Curtailment of reinforcement. Analysis and Design of sections in shear, bond and torsion, Diagonal tension, shear reinforcement, Development length, Anchorage and flexural bond, Torsional stiffness, equivalent shear, Torsional reinforcement, Design examples.

Module 4: Serviceability Limit State

Control of deflection, Cracking, Slenderness and vibrations, Deflection and moment relationship for limiting values of span to depth, Limit state of crack width, Design examples.

UNIT-C

Module 5: Slabs

General considerations, Design of one way and two ways slabs for distributed and concentrated loads, Non-rectangular slabs, Openings in slabs, Design Examples.

Module 6: Retaining Walls

Classification, Forces on retaining walls, Design criteria, Stability requirements, Proportioning of cantilever retaining walls, counter fort retaining walls, criteria for design of counter forts, Design examples.

UNIT-D

Module 7: Columns

Effective length, Minimum eccentricity, Short columns, under axial compression, Uniaxial and biaxial bending, Slender columns. Design examples.

Module 8: Footings

Isolated and wall footings, Design examples. Foundations-Combined footings, raft foundation, design of pile cap and piles, under reamed piles, design examples.

Course Outcomes:

At the end of the course, the students will be able to

- Recognize the design philosophy of reinforced concrete structures.
- Be able to analyze reinforced concrete structural systems under gravity and lateral loads.
- Be able to design different elements of reinforced concrete structural systems subjected to gravity and lateral loads.
- Be able to analyze and design a complete structural system through a comprehensive design project.

- Summarize the fundamental mechanics of reinforced concrete and the empirical assumptions made for analysis.
- Be able to produce a complete project document and present in a concise and complete manner to include structural drawings and structural calculations.
- Design basic structural elements (beams, columns and slabs) according to the design approach of IS:456.

SUGGESTED BOOKS:

- Design Of Reinforced Concrete Structures By P.Dayaratnam, Oxford & IBH Pub.,N.Delhi.
- Design of Reinforced Concrete-Limit State Design By A.K.Jain, Nem Chand & Bros.,Roorkee.
- Design of Reinforced Concrete by I.C.Syal & A,K,Goel, A.H,Wheeler & Co.Delhi.Reinforced Concrete Design by S.N.Sinha, Tmh Pub.,N.Delhi.
- Sp-16(S&T)-1980, Design Aids For Reinforced Concrete to IS:456, BIS, N.Delhi.
- Reinforced cement concrete design by Neelam Sharma , S.K.Kataria & sons, N.Delhi.
- Sp-34(S&T)-1987 Handbook on Concrete Reinforcement And Detailing`, BIS, N.Delhi.

STRUCTURAL ANALYSIS			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	3-0-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To provide basic knowledge in mechanics of materials so that the students can solve real engineering problems and Design Engineering Systems.
- Covers the relationship between stress and strain on deformable solids, principal stresses, maximum shearing stress, and the stresses acting on a structural member.
- Applies analysis to members subjected to axial, bending, and Torsional loads.
- Learn to evaluate internal forces, moments and corresponding stresses in beams through problem solving sessions using different methods.
- This course provides foundation knowledge, skills and their application which are relevant to subsequent courses in Civil Engineering.

COURSE CONTENT

UNIT-A

MODULE 1: Deflection of Statically determinate structures

Deflection of determinate beams by Double Integration Method, Conjugate Beam Method and Moment Area Methods, Principle of Virtual work (Unit load method) and Castiglano's theorem.

MODULE 2: Deflection of Statically determinate Frame & Truss

Deflection of determinate pin jointed trusses and rigid jointed frames by principle of virtual work, Strain Energy and Castiglino's theorem. Williot Mohr diagram method and Maxwell's laws of reciprocal theorem

UNIT-B

MODULE 3: Influence Line

Influence lines for reactions, BM & SF for simply supported beam and Panelled Girders. Influence lines for forces in trusses with top horizontal and curved both, Reversal of stresses, Use of influence lines for calculating design forces due to dead load and moving live loads. Influence lines using Muller Breslau principle.

UNIT-C

MODULE 4: Arches

Determination of horizontal thrust, shear force and bending moment diagram for:

1. Two Hinged Arches 2. Three Hinged Arches 3. Fixed Arches

MODULE 5: Column Analogy Method & Cable and Suspension Bridge

Elastic centre, properties of analogous column, application to beam & frames. Introduction of Cable and suspension Bridge uniformly loaded cables, Temperature stresses, and three hinged stiffening Girder and two hinged stiffening girder

UNIT-D

MODULE 6: Indeterminate Structures & Deflection methods

Introduction to Indeterminate Structures, Determination of kinematic and static indeterminacy of beams, frames and trusses, Slope Deflection and Moment Distribution Methods- Analysis of continuous beams & portal frames, Portal frames with inclined members.

Course Outcomes:

At the end of the course, the students will be able to:

- Helps to determine the deflections and rotations produced by the three fundamental types of loads: axial, Torsional, and flexural.
- Identify the internal forces and moments in beams to develop shear force and bending moment diagrams. Assess section properties, bending and deflections in beams.
- Use various classical methods for analysis of indeterminate structures.
- Determine the effect of support settlements for indeterminate structures.
- Apply the concepts of ILD and moving loads on structures.
- Demonstrate the concepts of qualitative influence line diagram for continuous beams and frames
- Apply the methods of indeterminate truss analysis demonstrate the behaviour of arches and their methods of analysis.

Suggested Books:

- Statically Indeterminate Structures by C.K. Wang, McGraw Hill Book Co., New York.
- Advanced Structural Analysis by A.K. Jain, Nem Chand & Bros., Roorkee.
- Indeterminate Structures by R.L. Jindal, S. Chand & Co., New Delhi.
- Theory of Structures, Vol. I, by S.P. Gupta & G.S.Pandit, Tata McGraw Hill, New Delhi.

SURVEYING-II			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	3-0-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To understand the principle of surveying on very large scale by locating precise horizontal controls.
- To learn about surveying applications in setting out works.
- To learn about determining absolute positions of a point using celestial measurements.
- To learn about different types of errors in measurements and their adjustment.
- To introduce the basic concept of photogrammetry, Remote sensing, and GIS.

COURSE CONTENT

UNIT A

Module 1: Curves

Classification of curves, elements of simple circular curve, location of tangent points- chain and tape methods, instrumental methods, Examples; types of transition curves; Vertical Curves: Necessity and types of vertical curves, setting out of a vertical curve by tangent correction, chord gradient and sight distance method.

UNIT B

Module 2: Survey Adjustment and computations

Definitions, types of error, weight of an observation, law of weights, most probable values, principle of least squares, method of correlates, normal equation, adjustment of triangulation figures by method of least squares.

UNIT C

Module 4: Elements of Photogrammetry

Introduction, types of photographs, aerial camera, scale of a photograph, height displacements of vertical photographs, flight planning and its uses, crab and drift, number of photographs, relief displacements, Stereoscopic vision and stereoscopes, height determination from parallax measurement, flight planning, principle of photo interpretation, photogrammetric monitoring.

UNIT D

Module 5: Introduction to remote sensing and Geographical Information System (GIS)

Definition of Remote Sensing, types of remote sensing, remote sensing system and components. EMR source and characteristics, active and passive remote sensing, EMR propagation through medium, EMR interaction with objects, Spectral signature, Satellite orbits and platforms: Geostationary and sun synchronous satellites, Applications of remote sensing in civil engineering.

Geographical Information System (GIS): Definition and Objectives, Components of GIS, Spatial data models: Raster and Vector.

Course Outcomes:

- Students would be able to know about advanced methods of locating horizontal controls.
- Set out various civil engineering structures, learn about different types of time and solution of astronomical triangle.
- Apply corrections to the measurements for different errors, and understand the difference between aerial photograph and satellite images and their use in map making.

Suggested Books:

- Chang.T.K. 2002: Geographic Information Systems, Tata McGrawHill
- Punmia, B.C. 2005: Surveying I and II, Luxmi Publications
- Charles D. Ghilani: Adjustment Computations: Spatial Data Analysis (Fifth Edition)
- Paul R Wolf: Elements of Photogrammetry
- G S Srivastava: An introduction to Geoinformatics
- Basudeb Bhatta: Remote Sensing and GIS
- G. L. Hosmer: Text-book on Practical Astronomy
- Various Online resources including NPTEL

Material Testing and Evaluation			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	3-0-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To provide the students an knowledge about various engineering materials.
- To understand the properties of ingredients of concrete.
- To study the behaviour of concrete under different states.
- To study about the concrete design mix.
- To understand special concrete and their use.
- To know various heavy construction projects and the equipments used for these.

COURSE CONTENT

UNIT-A

Module 1: Introduction to Engineering Materials

Cements, M-Sand, Concrete (plain, reinforced and steel fibre/glass fibre- reinforced, light-weight concrete, High Performance Concrete, Polymer Concrete) Ceramics and Refractories, Bitumen and asphaltic materials, Glass and Plastics, Structural Steel and other Metals

Module 2: Limes, cement and mortars

Lime: classification of lime, manufacturing, testing of lime, storage of lime, Cement: cements composition, types of cement, manufacturing of ordinary portland cement, special types of cement, storage of cement, testing of cement. Mortars: Proportions of lime and cement mortars, mortars for masonry and plastering.

UNIT-B

Module 3: Concrete making materials

Proportions of cements, aggregates, water and admixtures; properties of fresh and hardened concrete, variability of concrete strength, extreme weather concreting, prestressed concrete; Durability of concrete - alkali aggregate reaction, reinforcement corrosion, freezing and thawing, etc.

Module 4: Mix Design

Principles of concrete mix design, basic considerations, Factors in the choice of mix design, outline of mix design procedure, ACI mix design practice, USBR method, British mix design method IS guidelines.

UNIT-C

Module 5: Steel and its testing

Types of steel, mechanical behaviour and mechanical characteristics; Elasticity – principle and characteristics; Plastic deformation of metals; tensile test – standards for different material (brittle, quasi-brittle, elastic and so on); Bending and torsion test, procedure and standards, Strength of ceramic, Internal friction, creep – fundamentals and characteristics; Brittle fracture of steel – temperature transition approach; concept of fracture mechanics; fracture toughness testing.

UNIT-D

Module 6: Testing and Evaluation Procedures

Testing of concrete mixes, description for various concrete, steels, aggregates ; Elastic deformation; Plastic deformation; Impact test and transition temperatures; Fracture mechanics – background; Fracture toughness – different materials; Fatigue of material; Shrinkage, Creep.

Module 7: Construction equipments and Heavy Construction

Construction of large structures, dams, bridges, multi storeyed buildings etc, Construction Equipments - crushers, hot mix, plants, dozers etc, Introduction to heavy construction equipment.

Course Outcomes:

At the end of the course, the students will be able to

- To explain various type of constructions in Civil Engineering.
- Design the concrete mix using ACI and IS code methods.
- Determine the properties of fresh and hardened of concrete.
- Design special concretes and their specific applications ensure quality control while testing/ sampling and acceptance criteria.

SUGGESTED BOOKS:

- Handbook of mix design - BIS
- Concrete Technology by M.S. Shetty.
- Chudley, R., Greeno (2006), 'Building Construction Handbook' (6th ed.), R. Butterworth- Heinemann
- Khanna, S.K., Justo, C.E.G and Veeraragavan, A, ' Highway Materials and Pavement Testing', Nem Chand & Bros, Fifth Edition
- Various related updated & recent standards of BIS, IRC, ASTM, RILEM, AASHTO, etc. corresponding to materials used for Civil Engineering applications
- Kyriakos Komvopoulos (2011), Mechanical Testing of Engineering Materials, Cognella
- E.N. Dowling (1993), Mechanical Behaviour of Materials, Prentice Hall International Edition
- American Society for Testing and Materials (ASTM), Annual Book of ASTM Standards (post 2000)

FLUID MECHANICS-II LAB.		
Course Code		External marks: 50
Credits	1	Internal marks: 50
L-T-P	0-0-2	Total marks: 100

COURSE OBJECTIVES:

- To understand the flow measurement in a pipe flow.
- To determine the energy loss in pipe flow.
- To study the loss due to pipe fittings.
- To measure the discharge in a open channel flow etc.

LIST OF EXPERIMENTS

1. To determine the coefficient of drag by Stokes law for spherical bodies.
2. To study the phenomenon of cavitations in pipe flow.
3. To determine the critical Reynolds number for flow through commercial pipes.
4. To determine the coefficient of discharge for flow over a broad crested weir.
5. To study the characteristics of a hydraulic jump on a horizontal floor and sloping glacis including friction blocks.
6. To study the scouring phenomenon around a bridge pier model
7. To study the scouring phenomenon for flow past a spur.
8. To determine head loss due to various pipe fittings.

COURSE OUTCOMES:

At the end of the course, the students will be able to:

- Measure discharge in pipes determines the energy loss in conduits.
- Carry out discharge measurements in open channel etc.

STRUCTURAL ANALYSIS LAB			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- Structural Analysis experiments help to understand, to know the practical behaviour of the physical structures like beams, different arches, roof truss etc.
- A proper structural analysis of these structures helps the students to solve the practical problems.
- Different structural apparatus like Two-Hinge Arch, Three- Hinge Arch are studied in the laboratory.

COURSE CONTENT

UNIT-A

- 1 To verify moment area theorem regarding slope and deflection in a beam
- 2 To verify Maxwell's Reciprocal Theorem.
- 3 Begg`sdeformeter- verification of Muller Breslau principle
- 4 Experiment on a two – hinged arch for horizontal thrust and influence line for horizontal thrust
- 5 Analytical and experimental study of three hinged arch
- 6 Experimental and analytical study of unsymmetrical bending of a cantilever beam
- 7 Sway in portal frames – Demonstration

Course Outcomes:

At the end of the course, the students will be able to:

- Various experimental and analytical studies for different structural members and their comparison.
- Demonstration of frame.
- Able to calculate the about forces, moments and deflections.
- To understand the Able to calculate the deflection of springs
- To verify and compare different theoretical and experimental theorems.

SURVEYING-II LAB.			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To study and use of theodolite for angle measurements
- To use tacheometer for horizontal and vertical distances.
- To draw simple circular curves.
- To measure base line measurement.
- To study total station and its use for measuring distance, elevation and coordinates.

LIST OF EXPERIMENTS

1. Study various parts of a theodolite
2. Measurement of horizontal and vertical angles with theodolite
3. Measurement of Tachometric constants.
4. Calculating horizontal distance and elevations using tachometer.
5. Exercise of triangulation including base line measurement.
6. Setting out simple circular curves by deflection angle method.
7. Study the various parts of a total station.
8. Measurements of distance, elevation, coordinate with total station.

COURSE OUTCOMES:

On completion of the course, the students will be able to:

- Use the theodolite for measuring angles and use of tacheometer to determine distance and elevation.
- Draw simple circular curves.
- Calculate base line measurement and importance of triangulation.
- Use a total station to measure distance, elevation and coordinates.
- Use total station to plot a map of given area with software.

Material Testing & Evaluation Lab.			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To determine important properties of cement with different tests.
- To study the various test on aggregates and concrete.

LIST OF EXPERIMENTS

1. Standard consistency of cement using Vicat`s apparatus.
2. A) Fineness of cement by Sieve analysis and Blaine`s air permeability method.
B) Fineness modulus of coarse and fine aggregates.
3. Soundness of cement by Le-Chatelier`s apparatus.
4. Setting time of cement, initial and final of cement.
5. Compressive strength of cement.
6. A) Measurement of specific gravity of cement.
B) Measurement of Heat of Hydration of cement.
7. Moisture content and bulking of fine aggregate.
8. Workability of cement concrete by (a) Slump test (b) Compaction factor test (c) Flow table test.
9. Compressive strength of concrete by (a) Cube test, (b) Cylinder test
10. Indirect tensile strength of concrete-split cylinder test.

Course Outcomes:

At the end of the course, the students will be able to:

- To able understand the importance of testing of cement, sand and aggregate.
- Able to perform different tests of concrete to check their suitability.
- Study of various properties of cement, aggregate and concrete for any project work.
- To check the suitability of material for practical application.

RCC Drawing Lab.			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To understand the principles of planning and bylaws.
- To draw plan, elevation and section of RC Slabs, Beams, Columns and Footings and Portal Frame.
- To prepare detailed working design of a small RCC framed building

LIST OF EXPERIMENTS

RCC Drawing: Reinforcement details from the given data for the following structural elements with bar bending schedules

- (i) Drawing No. 1: RC Slabs - One way slab, Two way slab and Cantilever Slab.
- (ii) Drawing No.2 : Beams - Singly and doubly reinforced rectangular beams and Cantilever beam (All beams with vertical stirrups)
- (iii) Drawing No.3 : Columns and Footings – Square, Rectangular and Circular Columns with lateral ties and their isolated sloped column footings.
- (iv) Drawing No. 4 : Portal Frame – Three bay two storey RC portal frame with blow up of column beam junctions.
- (v) Drawing No. 5 : Design of a small RCC framed building using Limit State method of design including preparation of necessary working drawing.

INSTRUCTIONAL STRATEGY

Teachers are expected to give simple problems for designing various RCC structural members. For creating comprehension of the subject, teachers may prepare tutorial sheets, which may be given to the students for solving. It would be advantageous if students are taken at construction site to show form work for RCC as well as placement of reinforcement in various structural members. Commentary on BIS:456 may be referred along with code for relevant clauses.

Course Outcomes:

After going through this course, the students will be able to

Understand material properties and design methodologies for reinforced concrete structures.

- Assess different type of loads and prepare layout for reinforced concrete structures.
- Identify and apply the applicable industrial design codes relevant to the design of reinforced concrete members.
- Analyse and design various structural elements of reinforced concrete building like beam, slab, column, footing, and staircase.
- Assessment of serviceability criteria for reinforced concrete beam and slab.

- Prepare structural drawings and detailing and produce design calculations and drawing in appropriate professional format.